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Distraught Grapes - Final Report

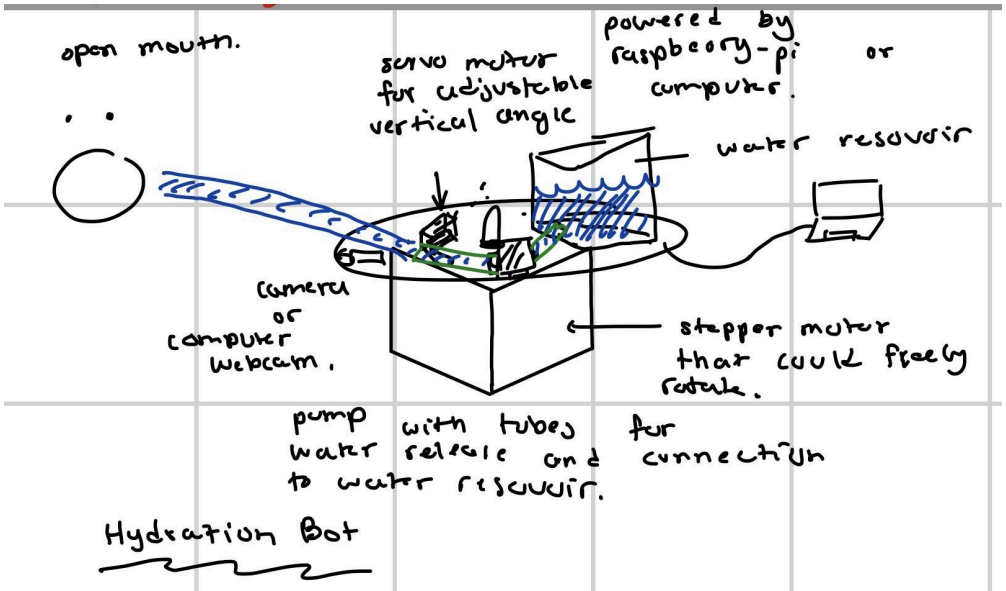
Introduction:

Our project, the “Hydration Bot”, is a hands-free automated hydration system. It solves the problem of delivering water without manual intervention by utilizing computer vision to track facial features and delivering precise, pulsed water streams directly into the user’s mouth when they open it. The system was designed for athletes in helmeted sports and individuals with limited mobility.

Our initial key design requirements were that our total cost of our project would need to be less than 200 dollars, our algorithm could detect an open mouth with 90 percent confidence, our target accuracy is within 10 millimeters at a distance of 1.5 meters, our mechanism being able to respond quickly within 150-300ms, and that we could deliver a stream of water within a safe psi range of approximately 5-10psi. Lastly, we wanted to run our program off of a Raspberry Pi or other similar microcomputer so it would be more compact and not require a laptop to run.

Prototype 1:

In our first prototype design, we first sketched a quick general layout of the main parts that our project would include. We would ideally have a pan tilt mechanism for the turret to aim, which our tubing, nozzle, and camera would be attached to. We would also keep this separate from our water reservoir and other electronics.

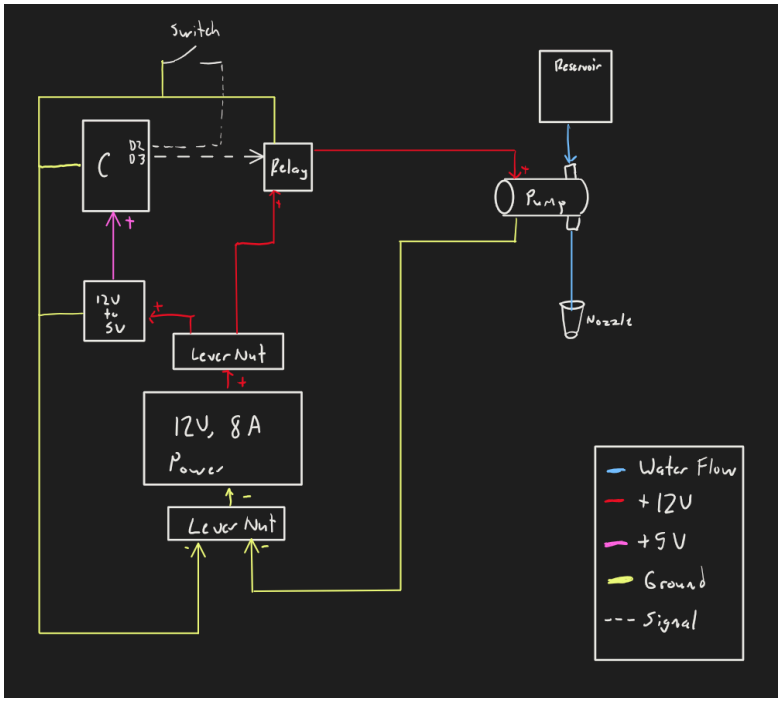


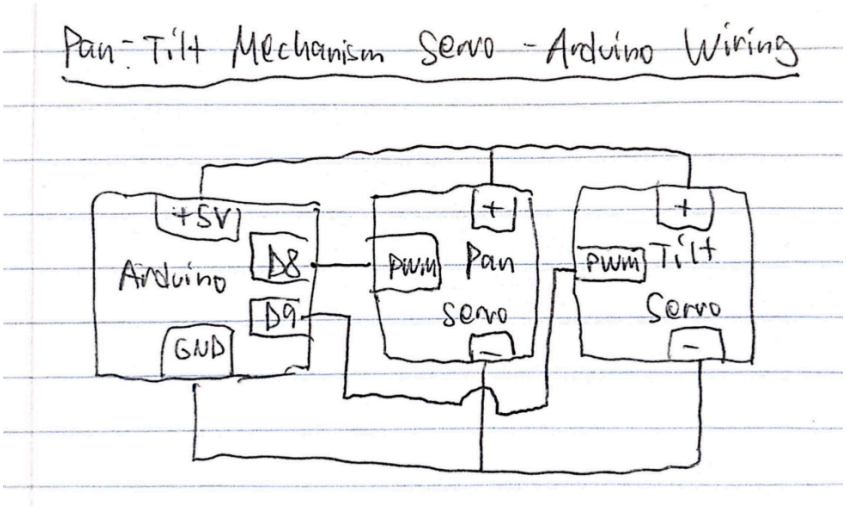
For our functioning parts, we took example code from google that used MediaPipe, an open source framework for computer vision and video face meshing, and edited it so that it would work with our webcam and we had our program window output if we were able to detect if the mouth is open while also outputting pixel coordinates for the position of the vertical center of the mouth. We were not able to get any plots for our data, but our program was able to detect an open mouth without fail, given that it had applied the facemesh to a person within frame and had locked on to that person. Additionally, we tested that it was only able to detect faces within 2 meters or less, otherwise our program cannot detect a face to apply the mesh on.

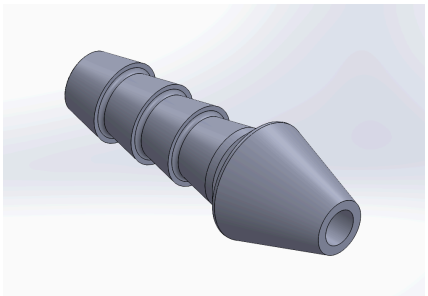
For our unknown, we were mainly concerned if the webcam we used would be able to detect an open mouth and provide accurate targeting coordinates in a reasonable amount of time since it is external hardware. With our prototype we were able to eliminate this concern, as it was easily able to hit our initial specification of our project being able to detect a mouth within 1.5 meters and also reacting within our target range of 150-300ms.

Prototype 2:

In our second prototype, we designed a nozzle to direct a stream of water and tested it with the pump that we selected. We created a system block diagram to show our general project flow and sketched a more detailed wiring diagram for our system. For our pump, we used a SEAFLO food grade 12 V DC diaphragm pump. We also used vinyl tubing to transfer water.







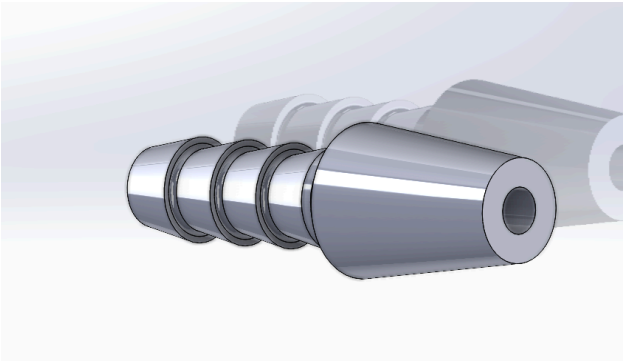
Our main concern/risk was whether or not the pump would not be able to deliver a clean stream of water in bursts so we could avoid spraying and getting the recipient wet. We tested our nozzle and tubing, aiming our stream into a bowl at varying distances from the outlet to see how our nozzle affected our spray and to see if we could maintain a consistent stream.

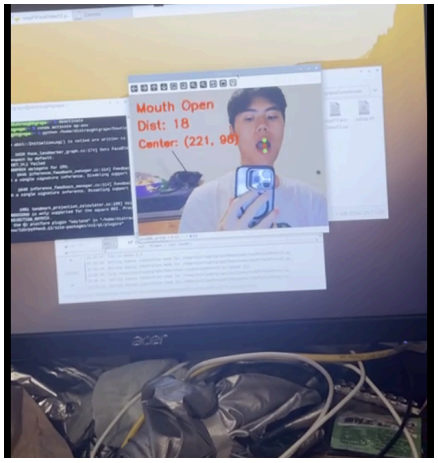


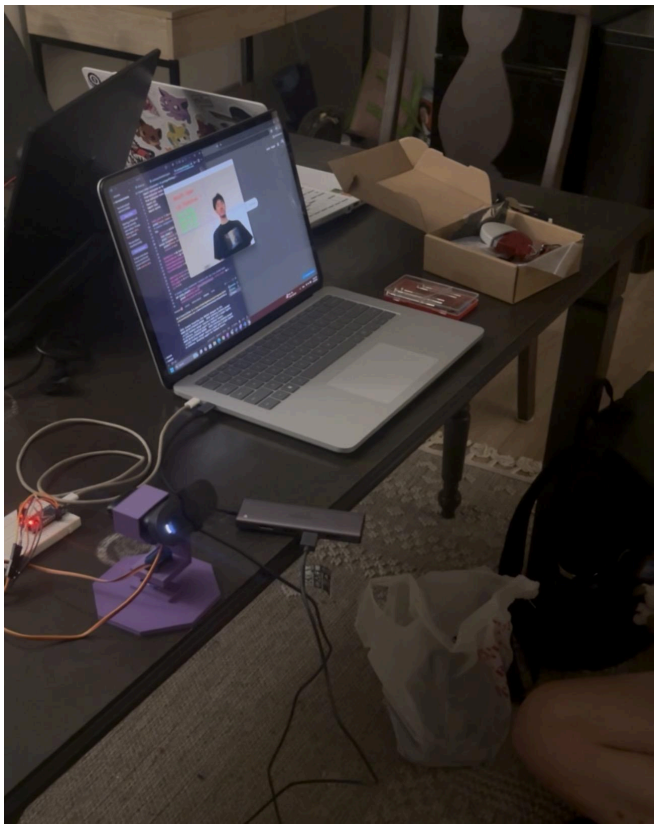
When testing, we measured our firing distance to be around 1.75 meters, which was comfortably over our 1.5 meter spec we initially listed.

Prototype 3:

In our third prototype, we redesigned our nozzle to reduce the amount of fraying in our spray, and we were able to integrate the mouth detection and face tracking with the pan tilt mechanism. Our pan-tilt mechanism worked with two SG90 servo motors, and we had one fixed in place horizontally, so its movement could rotate a disk on our x axis around 180 degrees, and we had the other motor fixed vertically, so its movement could pan another housing 90 degrees on our y axis to control aim. Additionally, we also tested and were able to successfully run our mouth detection code off of a webcam hooked up to a raspberry pi.





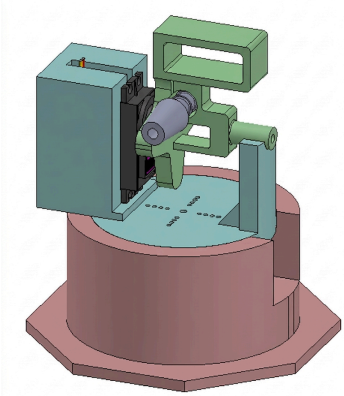


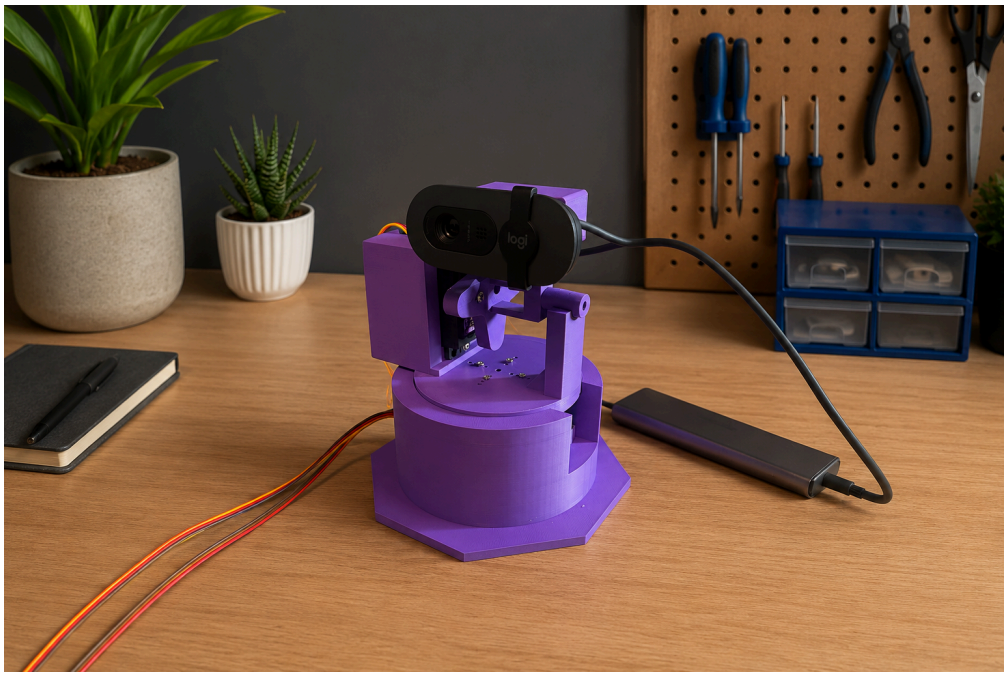
We attached our webcam to the pan tilt turret and got it to track our moving face with precision and trigger the pump once the camera centered itself on the person in the frame's open mouth. However, we were unable to get the pan tilt mechanism to move our tubing. Our biggest concern was if we would be able to get the pan tilt mechanism to aim with minimal jittering. When testing, the camera was having trouble settling on a center target. We decided to experiment with a derivative controller since derivative controllers are good for controlling overshoot. It helped but did not completely fix the problem. Our next step was to add a deadzone trigger, a small box region where the camera deems the center of the mouth that is larger than the initial coordinate we had. This fixed our unknown and allowed us to get closer to hitting our spec of accurately spraying water in a person’s mouth.

However, this pan-tilt mechanism still had some issues. We found that the pan-tilt mechanism was unable to properly bend the vinyl tubing that the nozzle was attached to. We also saw that the servo motors' shafts were not supported well, so the whole mechanism was wobbly and not as rigid as we had hoped. We began looking into solutions to these two issues for our final build.

Final Build:

In our final build we put all our components together to complete our system. We attached our tubing and nozzle to our pan tilt mechanism that housed our webcam and connected our tubing to our pump. We initially wanted to run our software off of a raspberry pi, however we ran into some issues regarding latency and the image recognition quality due to its weaker processing power. Due to this, we ran our final build using one of our laptops.





Additionally, in our final build we replaced our SG90 servos in our pan tilt mechanism with stronger MG996R servos that could output more torque, and we also replaced our vinyl

tubing with silicone tubing since it is more compliant. This required a redesign of the pan-tilt mechanism, since the MG996R servos were about twice the size of the previous SG90s. The new pan-tit mechanism (shown above) also had lateral supports for the tilt servo, so that the shaft that the webcam and nozzle were mounted to was no longer cantilevered. This fixed the wobbling issues we had observed in Prototype 3's pan-tilt.

Lastly, we ran into a new issue when testing the final build. We realized that when the relay tried to turn off the pump, the pump's inductance was causing a large voltage spike, which seemed to be causing issues with the Arduino - the Arduino wasn't actually able to command the relay to turn the pump off. To fix this, we added a diode in parallel to the pump in a flyback configuration: when a voltage spike occurred, current was able to flow through the diode and to the other side of the pump, quickly lowering the voltage to a safe amount.

With these upgrades (silicone tubing, stronger servos, new pan-tilt, and flyback diode), our project was able to hit most of our design specifications, save for not being able to run off of a raspberry pi/microcomputer and not being able to aim and shoot a distance of 1.5 meters. When running our final build, our tubing needed to be lengthened so it could reach our water reservoir, which created more head loss. This added head loss, combined with outside factors like wind, made it so that our project's maximum effective range was around 1.21 meters (4 feet) instead of our initial spec of 1.5 meters.

Other Evidence:

To help reduce uncertainty in our design, during our process we initially did some calculations to help us select our pump. This was all based around our specified distances.

- **Pump Flow Rate (Q):** 1.2 GPM $= 7.57 \times 10^{-5}$ m^3/s
- **Nozzle Diameter (d_n):** 4.67 mm
- **Nozzle Area (A_n):** 1.71×10^{-5} m^2

Fluid Dynamics

- **Exit Velocity (v_0):** $v_0 = \frac{Q}{A_n} = \frac{7.57 \times 10^{-5}}{1.71 \times 10^{-5}} \approx \mathbf{4.42}$ m/s
- **Pressure Drop (ΔP):** $\Delta P = \frac{1}{2}\rho v_0^2 = \frac{1}{2}(1000)(4.42)^2 \approx \mathbf{9768}$ Pa (1.41 PSI)
- *Note: 1.41 PSI is vastly below the 35 PSI pump limit, meaning flow is fully unrestricted.*

Ideal Projectile Kinematics ($\theta = 45°$)

- **Max Horizontal Range ($x_{\max}$):** $x_{\max} = \frac{v_0^2}{g} = \frac{4.42^2}{9.81} \approx \mathbf{1.99}$ m (6.5 ft)
- **Max Peak Height ($h_{\max}$):** $h_{\max} = \frac{v_0^2}{4g} = \frac{4.42^2}{39.24} \approx \mathbf{0.50}$ m (1.6 ft)

We also did some torque calculations to help us choose a more powerful motor after we tested in our third prototype that our initial SG90 servo motors were not strong enough to properly move around our tubing.

Pan-Tilt Requirements:

1. Angular acceleration: To accurately track a face: go from $\theta = 0$ rad to $\theta = \frac{\pi}{4}$ rad in 0.1 seconds
2. Moment of inertia: pan motor should be able to accelerate tilt mechanism, webcam, and nozzle: estimate weight of 1 kg and radius 5 cm

$$\theta = \frac{1}{2}\alpha t^2 \Rightarrow \alpha = \frac{2\theta}{t^2} = 157 \text{ rad / s}^2$$

$$I_{disk} = \frac{1}{2}MR^2 = \frac{1}{2}(1 \text{ kg})(0.05 \text{ m})^2 = 2.5 \times 10^{-3} \text{ kg m}^2$$

$$\tau_{pan} = I\alpha = (2.5 \times 10^{-3} \text{ kg m}^2)(157 \text{ rad / s}^2) = 0.39 \text{ Nm} = 4.0 \text{ kg cm}$$

Servo motor MG996R's stall torque:

$$\tau_{stall} = 9.4 \text{ kg cm}$$

"Factor of Safety" for torque:

$$FOS_{Torque} = \frac{\tau_{stall}}{\tau_{pan}} = \frac{9.4 \text{ kg cm}}{4.0 \text{ kg cm}}$$

$$\boxed{FOS_{Torque} = 2.35}$$

In the above calculation, the MG996R servo's stall torque was taken from the MG996R servo's datasheet.

Reflect:

1. What surprised you about the design process, about your team's product?

We were surprised at the unforeseen issues we ran into when we started integrating together the individual components of our project, like the issues of our tubing being too stiff or of the need for a flyback diode.

2. Where did you fail, where did you succeed?

We failed at our objective to shoot water at a range of 2 meters, with our max being at about 1.2 meters, and we failed at the objective of using a Raspberry Pi, using a computer instead. We succeeded in accurately and quickly tracking a user's face within the 1.2-meter range and shooting a burst of water into their mouth.

3. On reflection, do you feel that your prototype shows that your design has promise? – is it worth developing further?

This design does have promise, especially for outdoor sports where quick water delivery is useful. At the design fair, we received suggestions to improve the startup speed of the product by using a hydraulic accumulator to store pressured water instead of shooting directly from the pump each time. This would let the Hydration Bot be used for sports like long-distance cycling or running to shoot water into racers' mouths as they run by.

4. What changes would you make in the next iteration?

In the next iteration, we would use a hydraulic accumulator as suggested, and we would refine the nozzle mounting mechanism so that there is no uncertainty in how we install the nozzle into the device. This would make the device much more reliable in aiming, allowing us to calibrate better and further reduce the targeting error.

5. You will work on another design team in Capstone next year. Would you organize your team's work any differently looking back on this quarter's work?

We think our initial division of labor worked well for this project. The only thing we would do differently is try to integrate the individual components sooner, as the integration process came with many more problems than expected.